Jana Chamma

AI & ML Engineer · Computer Vision · Robotics Perception · Data Pipelines · Edge AI

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AI / ML Engineer	Data Scientist	Computer Vision Engineer	Backend ML Engineer	Robotics Engineer

AI & ML Engineer with 1.5 years of production experience building end-to-end machine learning systems, computer vision pipelines, and autonomous perception platforms. Expertise spans model deployment on edge hardware (NVIDIA Jetson), real-time data pipeline engineering, multi-sensor fusion, SLAM-based mapping, and cloud-integrated backend systems. Comfortable moving between research and production — from designing ML architectures to shipping APIs that serve them at scale. Open to relocation worldwide.

PROFESSIONAL EXPERIENCE

AI & Robotics Engineer

GlobalDWS · Toronto, Canada (Remote)

Feb 2025 – Present
Remote

- **[ML / CV]** Designed and deployed production ML inference pipelines on NVIDIA Jetson Nano/Orin combining YOLOv5 object detection, EasyOCR text extraction, and MINC material classification — running at real-time throughput using GStreamer NVMM camera input, CUDA acceleration, and a dual-threaded capture-inference architecture.
- **[ML / Data]** Built a semantic spatial understanding pipeline fusing LiDAR-generated occupancy maps with AI detections using pose-timestamp alignment — YOLO identifies fixtures, OCR extracts labels, MINC classifies materials — producing structured room-level JSON outputs used downstream in digital twin applications.
- **[Data Pipelines]** Engineered complete data pipelines from sensor acquisition through processing to cloud storage: structured telemetry ingestion, Azure Blob upload with retry logic and content-type routing, Azure SQL database insertion with deduplication logic, and CSV audit logging — handling high-frequency IoT data reliably at scale.
- **[Computer Vision]** Designed and documented a triple-sensor fusion workflow (LiDAR + Depth Camera + 360° Imagery) producing IFC-ready 3D building models via RealityCapture — combining Structure from Motion, Multi-View Stereo, and LiDAR geometry into textured reconstructions for digital twin and BIM applications (Doc: VIR-RealityCapture-003).
- **[Robotics / SLAM]** Deployed LiDAR-inertial SLAM pipelines using FastLIO + ROS2 with Livox MID360 and Pacecat M300 sensors — full stack from sensor driver configuration to 3D point cloud generation, PCD-to-LAS-to-COPC conversion via PDAL/Open3D, and 2D occupancy grid extraction.
- **[Backend / API]** Developed a FastAPI backend exposing the full autonomous pipeline via REST endpoints — background-threaded processing, persistent job state management, and artifact serving (maps, point clouds, room JSON) — demonstrating production-grade Python backend engineering.
- **[ML Systems]** Applied ML model optimization for edge deployment: evaluated TensorRT quantization, frame-rate budgeting, and keyframe-based inference scheduling to meet real-time constraints on Jetson hardware.

Stack: YOLOv5 · PyTorch · OpenCV · EasyOCR · MINC · FastLIO · ROS2 · PDAL · Open3D · FastAPI · Azure Blob · Azure SQL · Azure IoT · NVIDIA Jetson · TensorRT · CUDA · GStreamer · IFC · RealityCapture

AI Agent & Annotation Specialist

Aug 2025 – Present
Remote

Mindrift (Freelance)

- **[ML / AI]** Large-scale prompt engineering, LLM response evaluation for accuracy, ethics, clarity, and real-world user impact. Deepened expertise in human-AI interaction and responsible AI development.

Stack: Prompt Engineering · LLM Evaluation · AI Ethics · Annotation

IT Officer & Computer Trainer

Women Program Association (WPA)

Mar 2025 – Jun 2026
Lebanon

- Delivered digital literacy programs for 18–30 year old women. Provided hands-on IT support and drove operational digitization.

Stack: Digital Literacy · IT Support

STEM & Web Development Trainer

Al-Jana & Al-Smoud NGOs (Part-time)

Oct 2025 – Present
Lebanon

- Taught robotics, web development, and programming to youth. Designed 100+ hours of curriculum. Translating complex technical systems into accessible learning is a core engineering skill.

Stack: Robotics Education · Web Dev · Youth STEM

TECHNICAL SKILLS

ML & AI	PyTorch · TensorFlow · Scikit-learn · YOLOv5/v8 · CLIP · MINC · EasyOCR · Model Optimization · TensorRT · INT8 Quantization · Feature Engineering · Anomaly Detection
Computer Vision	OpenCV · Object Detection · Semantic Segmentation · 360° Perception · Quadrant Splitting · Material Classification · OCR · Real-time Video Processing · SfM · MVS · Photogrammetry
Data Engineering	Azure Blob Storage · Azure SQL · Azure IoT Central · AWS (S3, Lambda, IoT Core) · PostgreSQL · Pandas · CSV Pipelines · Telemetry Ingestion · Time-series Data · Data Deduplication
Robotics / SLAM	FastLIO · ROS2 · RViz · Gazebo · Nav2 · LiDAR-Inertial Odometry · Open3D · PDAL · COPC · Point Cloud Processing · Sensor Fusion · Occupancy Grids · Digital Twins · IFC / BIM
Edge / Embed	NVIDIA Jetson Nano/Orin · GStreamer · NVMM · CUDA · Real-time Inference · Embedded AI · Edge Deployment
Backend / Cloud	FastAPI · Flask · Docker · RabbitMQ · REST APIs · Microservices · CI/CD · JWT · GitLab · React.js · Next.js
Programming	Python · C++ · C · Java · Linux · Git
Languages	Arabic — Native · English — Fluent

SELECTED PROJECTS

01 · ML · EDGE AI **Real-Time 360° Perception Pipeline on Jetson**

Production ML inference system: YOLOv5 + EasyOCR + MINC on NVIDIA Jetson with dual-thread architecture, degree-based quadrant splitting for directional awareness, time-windowed deduplication, Azure SQL integration, and real-time telemetry upload. Optimized for constrained GPU memory and real-time throughput.

YOLOv5 · PyTorch · OpenCV · EasyOCR · MINC · Jetson Nano/Orin · GStreamer · Azure SQL · Azure Blob

02 · DATA PIPELINE · CLOUD **Autonomous Mapping Data Pipeline**

End-to-end data pipeline from LiDAR sensor acquisition to cloud storage: PCD → LAS → COPC conversion, structured telemetry ingestion, Azure Blob upload with retry/content-type logic, room-level JSON generation, and FastAPI REST backend with background-threaded processing and persistent job state.

FastAPI · PDAL · Open3D · Azure Blob · Azure SQL · Azure IoT · Python · Docker

03 · COMPUTER VISION · SENSOR FUSION **Triple-Sensor Fusion → IFC Digital Twin**

LiDAR + Depth Camera + 360° imagery fusion pipeline producing textured 3D reconstructions and IFC building models via RealityCapture (SfM + MVS). Combined with pose-anchored semantic enrichment using YOLO, OCR, and MINC to produce structured room metadata. Formally documented as R&D specification VIR-RealityCapture-003.

RealityCapture · LiDAR · Depth Camera · YOLO · OCR · MINC · IFC · BIM · SfM

04 · ML · PREDICTIVE ANALYTICS **Machine Failure Prediction**

ML classification model for detecting and categorizing machinery faults from sensor data. Explored feature engineering, anomaly detection, and model interpretability techniques for predictive maintenance applications.

Python · Scikit-learn · Pandas · Classification · Anomaly Detection · Feature Engineering

05 · AI · DATA ANALYSIS **Autonomous Systems Risk Analysis with Human-in-the-Loop**

Data-driven model analyzing risk and decision-making behavior in autonomous systems using real-world operational data. Implemented a human-in-the-loop validation framework. Combined ML with ethical evaluation to study bias, reliability, and failure modes in high-stakes environments.

Python · ML · Human-in-the-Loop · Data Analysis · AI Ethics

06 · BACKEND · DISTRIBUTED SYSTEMS **MelodySync — Microservices Media Pipeline**

Multi-service media conversion system: Docker containerization, async inter-service communication via RabbitMQ, JWT authentication, CI/CD pipelines with GitLab container registry.

Docker · RabbitMQ · Flask · Next.js · CI/CD · JWT · Microservices

EDUCATION

B.Sc. Computer Science — Graduated 2024

American University of Beirut (AUB)

USAID Full Scholarship · Dean's Honor List · AI, ML, systems, ethics, distributed computing

Exchange Semester — Machine Learning & AI Ethics — 2023

Carnegie Mellon University – Qatar (CMU-Q)

Education Above All Scholarship · Gulf region familiarity · Cross-cultural collaboration

CERTIFICATIONS & RECOGNITION

- Artificial Intelligence — ZAKA Academy
- Python for Data Science — ZAKA Academy
- Reality Capture Foundations for AEC — LinkedIn Learning
- Cisco ITE — Innovation Lab / UNICEF
- Aspire Leaders Program (Cohort 5) — Harvard-Affiliated
- Dean's Honor List — AUB
- Start for Good — Top 10 Finalist (CMU-Q / European Academy)
- AI Starter Kit — AUB